

Name: _____ (please print)

Signature: _____

ECE 2202 – Exam 3

November 8, 2025

**Keep this exam closed until you
are told to begin.**

1. This exam is closed book, closed notes. A formula sheet is provided in the back. Write your ID number at the top of every page.
2. Show all work on these pages. Show all work necessary to complete the problem. A solution without the appropriate work shown will receive no credit. A solution that is not given in a reasonable order will lose credit. Clearly indicate your answer (for example by enclosing it in a box).
3. Show all units in solutions, intermediate results, and figures. Units in the exam will be included between square brackets.
4. If the grader has difficulty following your work because it is messy or disorganized, you will lose credit.
5. Do not use red ink. Do not use red pencil.
6. You will have 100 minutes to work on this exam.

ID number: _____

GRADE

1. _____/25

2. _____/25

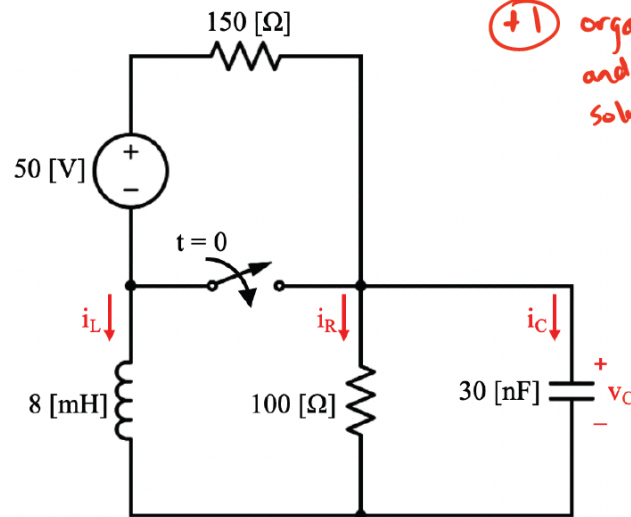
3. _____/25

4. _____/25

Total = 100

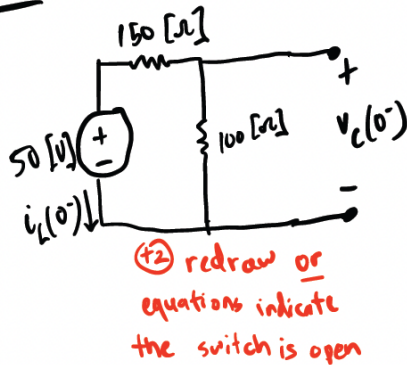
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1. {25 Points} The switch in the circuit below has been open for a long time before $t = 0$. Find $v_C(t)$ for $t \geq 0$.



(+1) organization - can read and follow work even if solution is incorrect

$t < 0$:



$$v_C(0^-) = 50[V] \cdot \left(\frac{100[\Omega]}{150 + 100[\Omega]} \right)$$

(+1) correct equation setup/value for $v_C(0^-)$

$$= 20[V]$$

$$i_L(0^-) = - \frac{50[V]}{150 + 100[\Omega]} = -200[mA]$$

(+1) correct equation setup/value for $i_L(0^-)$

$t \geq 0$: note - top part of circuit is shorted, so the 50[V] source and 150[Ω] resistor disappear



$$\alpha = \frac{1}{2RC} = \frac{1}{2 \cdot 100[\Omega] \cdot 30[nF]} = 166667[s^{-1}]$$

(+1) correct value α

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(8[mH])(30[nF])}} = 64549.7\left[\frac{rad}{s}\right]$$

(+1) correct value ω_0

$$\alpha > \omega_0 \Rightarrow \text{overdamping}$$

next page

Room for extra work

go to table 6-2:

$$v_c(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + v_c(\infty) \rightarrow 0$$

because no sources in the circuit due to the short

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -146667 [s^{-1}] + \sqrt{(146667)^2 - (64549.7)^2} = -13007.6 [s^{-1}]$$

(+1) correct s_1 value

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -146667 [s^{-1}] - \sqrt{(146667)^2 - (64549.7)^2} = -320326 [s^{-1}]$$

(+1) correct s_2 value

$$v_c(t) = A_1 e^{-13007.6t} + A_2 e^{-320326t}$$

① $v_c(0) = 20 [V] = A_1 + A_2 \Rightarrow$ one equation for two unknowns, need one more

KCL @ a $i_c + i_R + i_L = 0 \Rightarrow i_c(0) = -i_L(0) - i_R(0)$

(+2) correct KCL

$$i_c(0) = -(-0.2 [A]) - \frac{20 [V]}{100 [\Omega]} = 0 [A]$$

(+2) correct $i_c(0)$ value

$$i_c = C \frac{dv_c(t)}{dt} \Rightarrow \frac{i_c(0)}{C} = \frac{dv_c(0)}{dt} = A_1 s_1 + A_2 s_2$$

$$\frac{0 [A]}{30 [nF]} = \frac{dv_c(0)}{dt} = -13007.6 A_1 - 320326 A_2 = 0 [V] \quad \text{②}$$

① $20 [V] = A_1 + A_2$

② $0 [V] = -13007.6 A_1 - 320326 A_2$

} system of equations (+4) correct system of equations

$A_1 = 20.85 [V], A_2 = -0.85 [V]$ (+2) A_1, A_2 values correct

$$v_c(t) = 20.85 e^{-13007.6t} - 0.85 e^{-320326t} [V] \text{ for } t \geq 0$$

(+3) final $v_c(t)$ equation

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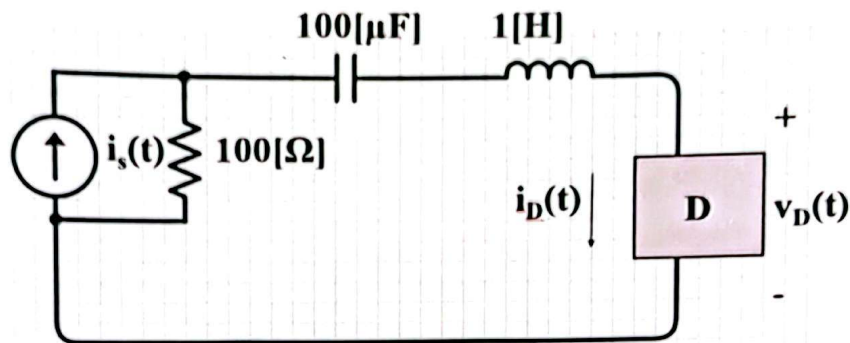
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2. (25 Points) The circuit below is operating in steady-state with $i_s(t)$ and $v_D(t)$ as given below.

A) Find current $i_D(t)$.

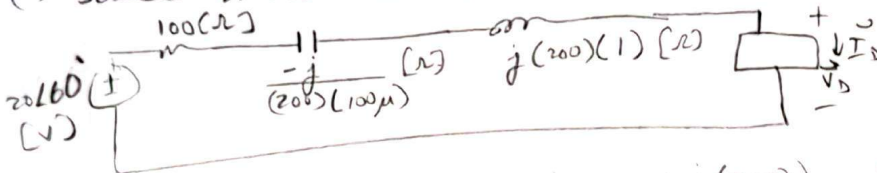
B) Find the device model of D. Assume D is a passive device. Draw the model of the device.



$$i_s(t) = 0.2 \cos\left(200 \left[\frac{\text{rad}}{\text{s}}\right] t + 60^\circ\right) [\text{A}] \Leftrightarrow 0.2 \angle 60^\circ [\text{A}] = \vec{I}_s$$

$$v_D(t) = 240 \cos\left(200 \left[\frac{\text{rad}}{\text{s}}\right] t + 45^\circ\right) [\text{V}] \Leftrightarrow 240 \angle 45^\circ [\text{V}] = \vec{V}_D$$

(A) Source transformation:



KVL eqn:

$$-20 \angle 60^\circ [\text{V}] + \vec{I}_D \left(100 - \frac{j}{(200)(100\mu)} + j(200) \right) + 240 \angle 45^\circ [\text{V}] = 0$$

$$\vec{I}_D (100 + j150) = -240 \angle 45^\circ + 20 \angle 60^\circ = -159.7056 - j152.3851$$

$$\vec{I}_D = -1.1947 + j0.2682 = 1.2245 \angle 167.3464^\circ$$

$$i_D(t) = 1.2245 \cos\left(200 \left[\frac{\text{rad}}{\text{s}}\right] t + 167.3464^\circ\right) [\text{A}]$$

(B) $\vec{Z}_D = \frac{\vec{V}_D}{\vec{I}_D} = \frac{240 \angle 45^\circ}{1.2245 \angle 167.3464^\circ} = 196.0052 \angle -122.3464^\circ [\Omega]$

$$\frac{1}{200 \left[\frac{\text{rad}}{\text{s}}\right] C} = 165.5909$$

$$C = 30.195 [\mu\text{F}]$$

$\vec{Z}_D$:

3. (25 Points) For the circuit below, assume the following values:
For $t \geq 0$, $v_{in} = 5[V]$, $R_s = 3[k\Omega]$, $R_f = 47[k\Omega]$, $C_1 = C_2 = 1[\mu F]$.

A) Find v_{C1} and v_{C2} for $t < 0$. Assume $v_{in} = 0[V]$ for $t < 0$.

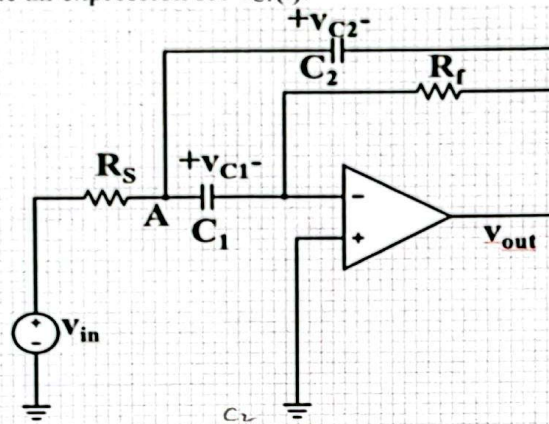
B) Find $v_{C1}'(0)$ (i.e., $dv_{C1}(0)/dt$)

C) Find $v_{C1}(\infty)$

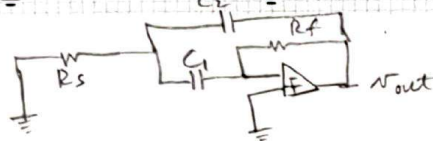
D) Starting with a KCL at node A, write a 2nd order differential equation in variable v_{C1} .

E) Determine if the response is overdamped, underdamped, or critically damped.

F) Write an expression for $v_{C1}(t)$.



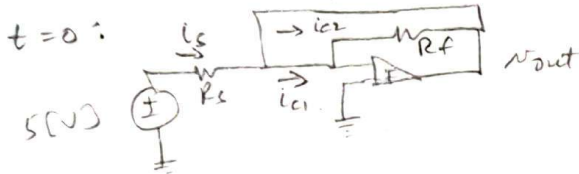
(A) $t < 0$:



$$v_{C1}(0^-) = 0[V]$$

$$v_{C2}(0^-) = 0[V]$$

(B) if $v_{C1}(0^-) = v_{C2}(0^-) = 0[V]$, then at $t = 0$ $v_{C1}(0) = v_{C2}(0) = 0$
caps act like shorts.



$$\frac{dv_{C1}(t)}{dt} = \frac{i_{C1}}{C_1}$$

$i_s = i_{C1} + i_{C2}$, but $i_{C1} = 0$ due to R_f being path of higher resistance

$$\text{so } \frac{dv_{C1}(0)}{dt} = 0[V/s]$$

(C) $t = \infty$: caps are open,

$$v_{C1}(\infty) = v_{in} = 5[V]$$

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Room for extra work

$$(D) \frac{v_{C1} - v_{in}}{R_s} + C_1 \frac{dv_{C1}}{dt} + C_2 \frac{d(v_{C1} - v_{out})}{dt} = 0$$

represent v_{out} in terms of v_{C1} :

$$C_1 \frac{dv_{C1}}{dt} = \frac{(0 - v_{out})}{R_f}$$

$$-R_f C_1 \frac{dv_{C1}}{dt} = v_{out}$$

$$\frac{v_{C1}}{R_s} + \frac{v_{in}}{R_s} + C_1 v_{C1}' + C_2 (v_{C1} + R_f C_1 v_{C1}')' = 0$$

$$v_{C1}'' (R_s C_1 C_2) + v_{C1}' (C_1 + C_2) + v_{C1} \left(\frac{1}{R_s} \right) = \frac{-v_{in}}{R_s}$$

$$(0.047 \times 10^{-6}) v_{C1}'' + (2 \times 10^{-6}) v_{C1}' + (333.33 \times 10^{-6}) v_{C1} = -0.0017$$

get into form $x'' + ax' + bx = c$:

$$v_{C1}'' + 42.5532 v_{C1}' + 7092.2 v_{C1} = -3617.0$$

$$\alpha = \frac{a}{2} = 21.2766 \text{ (Nepers/s)}$$

$$\omega_0 = \sqrt{b} = 84.2152 \text{ (rad/s)}$$

 $\omega_0 > \alpha$ underdamped

$$(E) D_1 = v_{C1}(0) - v_{C1}(\infty) = 0 - 5 \text{ [V]} = -5 \text{ [V]}$$

$$D_2 = \frac{v_{C1}'(0) + \alpha (v_{C1}(0) - v_{C1}(\infty))}{\sqrt{\omega_0^2 - \alpha^2}} = \frac{0 + 21.2766(-5)}{81.4832}$$

$$= -1.3056$$

$$v_{C1}(t) = (-5 \cos 81.4832t - 1.3056 \sin 81.4832t) e^{-21.2766t} + 5 \text{ [V]}$$

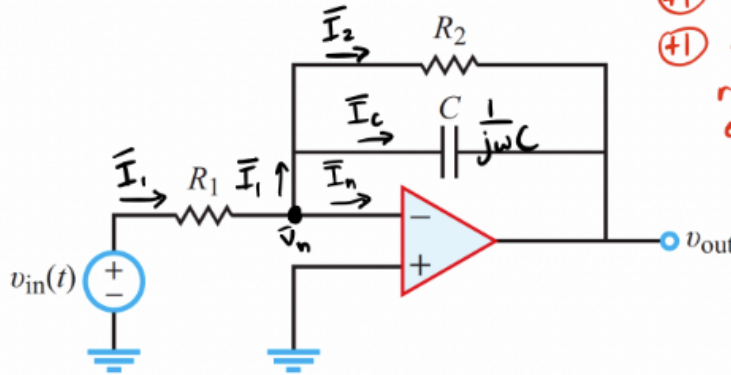
for $t \geq 0$

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4. {25 Points} The input signal in the op-amp circuit below is given by:

$$v_{in}(t) = 0.1 \cos(4500t) \text{ [V]}$$

Obtain an expression for $v_{out}(t)$ and then evaluate it for $R_1 = 5 \text{ [k}\Omega\text{]}$, $R_2 = 15 \text{ [k}\Omega\text{]}$, and $C = 0.5 \text{ [}\mu\text{F}\text{]}$. Solve this problem **only in the phasor domain** and then provide the final answer in the time domain.



(+1) labeling currents
(+1) organization - can read and follow work even if solution is incorrect

$$v_{in}(t) = 0.1 \cos 4500t \text{ [V]} \Rightarrow \bar{V}_{in} = 0.1 \angle 0^\circ \quad (+2) \text{ polar form of } V_{in}(t)$$

assumptions: $\bar{I}_n = 0$, $\bar{V}_n = 0$ (+2) ^{assumptions} explicitly written or assumed in the following math

KCL @ node $\bar{V}_n$ $\bar{I}_1 = \bar{I}_2 + \bar{I}_c$

(+5) KCL equation initial form is similar $\rightarrow$

$$\frac{\bar{V}_{in}}{R_1} = \frac{\bar{V}_n - \bar{V}_{out}}{R_2} + \frac{\bar{V}_n - \bar{V}_{out}}{1/j\omega C}$$

$$\frac{\bar{V}_{in}}{R_1} = \frac{-\bar{V}_{out}}{R_2} - j\omega C \bar{V}_{out}$$

$$\frac{\bar{V}_{in}}{R_1} = -\bar{V}_{out} \left(\frac{1}{R_2} + j\omega C \right)$$

$$-\frac{\bar{V}_{in}}{R_1} \cdot \left(\frac{1}{\frac{1}{R_2} + j\omega C} \right) = \bar{V}_{out}$$

Complex conjugate

$$\frac{(1-j\omega R_2 C)}{(1-j\omega R_2 C)} \left[-\frac{\bar{V}_{in}}{R_1} \cdot \left(\frac{1}{\left(\frac{1}{R_2} \right) (1+j\omega R_2 C)} \right) \right] = \bar{V}_{out}$$

$$-\bar{V}_{in} \cdot \frac{R_2}{R_1} \cdot \left(\frac{1-j\omega R_2 C}{(1-j\omega R_2 C)(1+j\omega R_2 C)} \right) = \bar{V}_{out}$$

$$-\bar{V}_{in} \cdot \frac{R_2}{R_1} \cdot \frac{1-j\omega R_2 C}{j\omega R_2 C - j\omega R_2 C + 1 + (-j^2 \omega^2 R_2^2 C^2)} = \bar{V}_{out}$$

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Room for extra work

$$\bar{V}_{out} = -\left(\frac{R_2}{R_1}\right) \left(\frac{1 - j\omega R_2 C}{1 + \omega^2 R_2^2 C^2}\right) \bar{V}_{in}$$

⊕4 getting to this form with or without all the math above

$$\bar{V}_{out} = -\left(\frac{15 [k\Omega]}{5 [k\Omega]}\right) \left(\frac{1 - j \cdot 4500 [\text{rad/s}] \cdot 15 [k\Omega] \cdot 0.5 [\mu F]}{1 + (4500 [\text{rad/s}])^2 (15 [k\Omega])^2 (0.5 [\mu F])^2}\right) \cdot 0.1 \angle 0^\circ$$

$$\bar{V}_{out} = -0.263 + j8.98 [\text{mV}]$$

$$= 8.89 e^{j91.7^\circ} [\text{mV}]$$

⊕4 either form correct before converting to time domain

$$v_{out}(t) = 8.89 \cos(4500t + 91.7^\circ) [\text{mV}]$$

⊕6 correct expression for $v_{out}(t)$

$$i = \frac{dq}{dt} \quad v = \frac{dw}{dq} \quad p = \frac{dw}{dt} = \frac{dw}{dq} \frac{dq}{dt} = iv$$

$$i = C \frac{dv}{dt} \quad v = L \frac{di}{dt}$$

$$v = iR$$

$$R_1 || R_2 = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{R_1 R_2}{R_1 + R_2}$$

$$p = Cv \frac{dv}{dt} \quad p = Li \frac{di}{dt}$$

$$p = iv = i^2 R = \frac{v^2}{R}$$

$$w = \frac{1}{2} Li^2 \quad w = \frac{1}{2} Cv^2$$

$$R = \frac{v}{i}$$

$$v_C(t) = v_C(\infty) + [v_C(T_0) - v_C(\infty)] e^{-(t-T_0)/\tau}$$

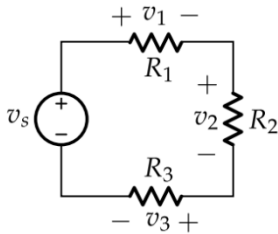
$$i_L(t) = i_L(\infty) + [i_L(T_0) - i_L(\infty)] e^{-(t-T_0)/\tau}$$

(Switch action at $t = T_0$)

$$\text{ODE: } \frac{dx}{dt} + ax = b, \text{ General solution: } x(t) = x(0)e^{-at} + \frac{b}{a}(1 - e^{-at})$$

$$\text{ODE: } \frac{dx}{dt} + ax = 0, \text{ General solution: } x(t) = x(0)e^{-at}$$

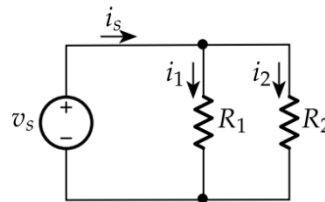
$$\text{Note that: } \frac{b}{a} = x(\infty), a = \frac{1}{\tau}$$



$$v_1 = \frac{R_1}{R_1 + R_2 + R_3} v_s$$

$$v_2 = \frac{R_2}{R_1 + R_2 + R_3} v_s$$

$$v_3 = \frac{R_3}{R_1 + R_2 + R_3} v_s$$



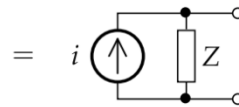
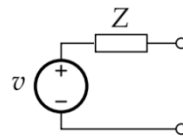
$$i_1 = \frac{R_2}{R_1 + R_2} i_s$$

$$i_2 = \frac{R_1}{R_1 + R_2} i_s$$

$$v_B = \underbrace{V_B}_{\text{DC}} + \underbrace{v_b}_{\text{AC}}$$

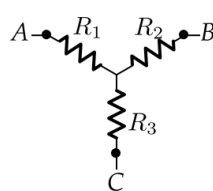
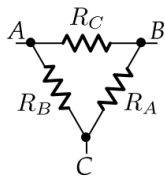
$\bar{V}_b \rightarrow$ Phasor notation

$$\text{CMRR} = 20 \log_{10} \frac{|A_d|}{|A_{cm}|}$$



$$v = i \cdot Z$$

$$i = \frac{v}{Z}$$



$$R_1 = \frac{R_B R_C}{R_A + R_B + R_C}$$

$$R_2 = \frac{R_A R_C}{R_A + R_B + R_C}$$

$$R_3 = \frac{R_A R_B}{R_A + R_B + R_C}$$

$$R_A = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$$

$$R_B = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2}$$

$$R_C = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_3}$$

Table 6-2: General solution for second-order circuits for $t \geq 0$.

$x(t)$ = unknown variable (voltage or current)	
Differential equation:	$x'' + ax' + bx = c$
Initial conditions:	$x(0)$ and $x'(0)$
Final condition:	$x(\infty) = \frac{c}{b}$
$\alpha = \frac{a}{2}$	$\omega_0 = \sqrt{b}$
Overdamped Response $\alpha > \omega_0$	
$x(t) = [A_1 e^{s_1 t} + A_2 e^{s_2 t} + x(\infty)] u(t)$	
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$
$A_1 = \frac{x'(0) - s_2[x(0) - x(\infty)]}{s_1 - s_2}$	$A_2 = -\left[\frac{x'(0) - s_1[x(0) - x(\infty)]}{s_1 - s_2} \right]$
Critically Damped $\alpha = \omega_0$	
$x(t) = [(B_1 + B_2 t)e^{-\alpha t} + x(\infty)] u(t)$	
$B_1 = x(0) - x(\infty)$	$B_2 = x'(0) + \alpha[x(0) - x(\infty)]$
Underdamped $\alpha < \omega_0$	
$x(t) = [(D_1 \cos \omega_d t + D_2 \sin \omega_d t)e^{-\alpha t} + x(\infty)] u(t)$	
$D_1 = x(0) - x(\infty)$	$D_2 = \frac{x'(0) + \alpha[x(0) - x(\infty)]}{\omega_d}$
$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$	
Auxiliary Relations	
$\alpha = \begin{cases} \frac{R}{2L} & \text{Series RLC} \\ \frac{1}{2RC} & \text{Parallel RLC} \end{cases}$	$\omega_0 = \frac{1}{\sqrt{LC}}$
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$
	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$

$$\vec{S} = \overline{V_{rms} I_{rms}^*} = |\overline{I_{rms}}|^2 \vec{Z} = P + jQ$$

$$P = \frac{V_m I_m}{2} \cos \theta$$

$$Q = \frac{V_m I_m}{2} \sin \theta$$

$$pf = \cos \theta$$

$$rf = \sin \theta$$